

HDOC DAY 17 : ASSIGNMENT

AIM: To write a menu-driven C program using a switch-case statement to perform various digit-based operations (generating the largest/smallest number from digits, swapping first and last digits, and checking for palindrome) on a positive integer using void user-defined functions without passing parameters.

ALGORITHM:

Main Program:

1. Start the program.
2. Display the main menu options to the user:
 - a. 1: Largest Number
 - b. 2: Smallest Number
 - c. 3: Swap First and Last Digit
 - d. 4: Palindrome Check
 - e. 5: Exit
3. Read the user's choice.
4. Using switch(choice):
 - a. Case 1: Call findLargest() function.
 - b. Case 2: Call findSmallest() function.
 - c. Case 3: Call swapFirstLast() function.
 - d. Case 4: Call checkPalindrome() function.
 - e. Case 5: Exit the program.
 - f. Default: Print "Invalid Choice".
5. Stop.

Function 1: findLargest()

1. Read an integer num.
2. Extract all digits and store frequency of each digit (0 to 9) in an array.

3. Print digits in descending order from 9 down to 0 based on their frequency.

Function 2: findSmallest()

1. Read an integer num.
2. Extract all digits and store frequency of each digit (0 to 9) in an array.
3. Print the smallest non-zero digit first (to avoid leading zero).
4. Print remaining digits in ascending order from 0 to 9 based on their frequency.

Function 3: swapFirstLast()

1. Read an integer num.
2. Extract the last digit using $\text{num} \% 10$.
3. Count total number of digits and find the first digit by repeatedly dividing num by 10.
4. Form the new number by swapping the first and last digits using arithmetic operations.
5. Print the swapped number.

Function 4: checkPalindrome()

1. Read an integer num.
2. Store num in a temporary variable temp.
3. Reverse the number by repeatedly finding digits ($\text{temp} \% 10$) and building $\text{rev} = \text{rev} * 10 + \text{digit}$.
4. If original num equals rev, print "Palindrome".
5. Else, print "Not Palindrome".

C PROGRAM CODE:

```
#include <stdio.h>
```

```
#include <math.h>

// Function Declarations

void findLargest();

void findSmallest();

void swapFirstLast();

void checkPalindrome();

int main() {
    int choice;

    printf("\n--- DIGIT OPERATIONS MENU ---\n");
    printf("1. Generate Largest Number\n");
    printf("2. Generate Smallest Number\n");
    printf("3. Swap First and Last Digit\n");
    printf("4. Check Palindrome\n");
    printf("5. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            findLargest();
            break;
```

```

    case 2:
        findSmallest();
        break;
    case 3:
        swapFirstLast();
        break;
    case 4:
        checkPalindrome();
        break;
    case 5:
        printf("Exiting program.\n");
        break;
    default:
        printf("Invalid choice!\n");
}

return 0;
}

// 1. Function to find largest number from digits
void findLargest() {
    int num, temp, count[10] = {0};
    printf("\nEnter a positive integer: ");

```

```
scanf("%d", &num);

temp = num;
while (temp > 0) {
    int rem = temp % 10;
    count[rem]++;
    temp = temp / 10;
}

printf("Largest possible number: ");
for (int i = 9; i >= 0; i--) {
    while (count[i] > 0) {
        printf("%d", i);
        count[i]--;
    }
}

printf("\n");
}
```

// 2. Function to find smallest number from digits

```
void findSmallest() {
    int num, temp, count[10] = {0};
    printf("\nEnter a positive integer: ");
```

```
scanf("%d", &num);
```

```
temp = num;
```

```
while (temp > 0) {
```

```
    int rem = temp % 10;
```

```
    count[rem]++;
```

```
    temp = temp / 10;
```

```
}
```

```
printf("Smallest possible number: ");
```

```
// Print first non-zero digit
```

```
for (int i = 1; i <= 9; i++) {
```

```
    if (count[i] > 0) {
```

```
        printf("%d", i);
```

```
        count[i]--;
```

```
        break;
```

```
    }
```

```
}
```

```
// Print remaining digits including 0s
```

```
for (int i = 0; i <= 9; i++) {
```

```
    while (count[i] > 0) {
```

```
        printf("%d", i);
```

```
        count[i]--;  
    }  
}  
printf("\n");  
}
```

// 3. Function to swap first and last digit

```
void swapFirstLast() {  
    int num, firstDigit, lastDigit, digits = 0, temp, swappedNum;  
    printf("\nEnter a positive integer: ");  
    scanf("%d", &num);  
  
    temp = num;  
    lastDigit = temp % 10;  
  
    while (temp >= 10) {  
        temp = temp / 10;  
        digits++;  
    }  
    firstDigit = temp;  
  
    if (num < 10) {  
        printf("Swapped number: %d\n", num);  
        return;  
    }  
}
```

```
}
```

```
int powVal = pow(10, digits);
```

```
swappedNum = lastDigit * powVal;
```

```
swappedNum += num % powVal;
```

```
swappedNum -= lastDigit;
```

```
swappedNum += firstDigit;
```

```
printf("Swapped number: %d\n", swappedNum);
```

```
}
```

```
// 4. Function to check if palindrome
```

```
void checkPalindrome() {
```

```
    int num, temp, rev = 0, rem;
```

```
    printf("\nEnter a positive integer: ");
```

```
    scanf("%d", &num);
```

```
    temp = num;
```

```
    while (temp > 0) {
```

```
        rem = temp % 10;
```

```
        rev = rev * 10 + rem;
```

```
        temp = temp / 10;
```

```
}
```



```

if (num == rev) {
    printf("%d is a Palindrome number.\n", num);
} else {
    printf("%d is NOT a Palindrome number.\n", num);
}
}

```

TEST CASE TABLE :

Test case no	Menu option	input	Expected Output	Actual Output	Status
1.	1	5824	8542	8542	pass
2.	2	5082	2058	2058	pass
3.	3	7394	4397	4397	pass
4.	4	12321	12321 is a Palindrome number.	12321 is a Palindrome number.	pass

3CODE SCREENSHOT AND EXECUTION ;

```
#include <stdio.h>
#include <math.h>

// Function Declarations
void findLargest();
void findSmallest();
void swapFirstLast();
void checkPalindrome();

int main() {
    int choice;

    printf("\n--- DIGIT OPERATIONS MENU ---\n");
    printf("1. Generate Largest Number\n");
    printf("2. Generate Smallest Number\n");
    printf("3. Swap First and Last Digit\n");
    printf("4. Check Palindrome\n");
    printf("5. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            findLargest();
            break;
        case 2:
            findSmallest();
            break;
        case 3:
            swapFirstLast();
            break;
        case 4:
            checkPalindrome();
            break;
        case 5:
            printf("Exiting program.\n");
            break;
        default:
            printf("Invalid choice!\n");
    }

    return 0;
}

// 1. Function to find largest number from digits
void findLargest() {
    int num, temp, count[10] = {0};
    printf("\nEnter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    while (temp > 0) {
        int rem = temp % 10;
        count[rem]++;
        temp = temp / 10;
    }
}
```

```

printf("Largest possible number: ");
for (int i = 9; i >= 0; i--) {
    while (count[i] > 0) {
        printf("%d", i);
        count[i]--;
    }
}
printf("\n");
}

// 2. Function to find smallest number from digits
void findSmallest() {
    int num, temp, count[10] = {0};
    printf("\nEnter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    while (temp > 0) {
        int rem = temp % 10;
        count[rem]++;
        temp = temp / 10;
    }

    printf("Smallest possible number: ");

    // Print first non-zero digit
    for (int i = 1; i <= 9; i++) {
        if (count[i] > 0) {
            printf("%d", i);
            count[i]--;
            break;
        }
    }

    // Print remaining digits including 0s
    for (int i = 0; i <= 9; i++) {
        while (count[i] > 0) {
            printf("%d", i);
            count[i]--;
        }
    }
    printf("\n");
}

// 3. Function to swap first and last digit
void swapFirstLast() {
    int num, firstDigit, lastDigit, digits = 0, temp, swappedNum;
    printf("\nEnter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    lastDigit = temp % 10;

```

```

while (temp >= 10) {
    temp = temp / 10;
    digits++;
}
firstDigit = temp;

if (num < 10) {
    printf("Swapped number: %d\n", num);
    return;
}

int powVal = pow(10, digits);
swappedNum = lastDigit * powVal;
swappedNum += num % powVal;
swappedNum -= lastDigit;
swappedNum += firstDigit;

printf("Swapped number: %d\n", swappedNum);
}

// 4. Function to check if palindrome
void checkPalindrome() {
    int num, temp, rev = 0, rem;
    printf("\nEnter a positive integer: ");
    scanf("%d", &num);

```

```
// 4. Function to check if palindrome
void checkPalindrome() {
    int num, temp, rev = 0, rem;
    printf("\nEnter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    while (temp > 0) {
        rem = temp % 10;
        rev = rev * 10 + rem;
        temp = temp / 10;
    }

    if (num == rev) {
        printf("%d is a Palindrome number.\n", num);
    } else {
        printf("%d is NOT a Palindrome number.\n", num);
    }
}
```

OUTPUT SCREENSHOT :

```
[manalithakar@Manalis-MacBook-Air ~ % nano 17q1.c
[manalithakar@Manalis-MacBook-Air ~ % gcc 17q1.c -o 17q1
[manalithakar@Manalis-MacBook-Air ~ % ./17q1
```

--- DIGIT OPERATIONS MENU ---

1. Generate Largest Number
2. Generate Smallest Number
3. Swap First and Last Digit
4. Check Palindrome
5. Exit

Enter your choice: 1

Enter a positive integer: 5824

Largest possible number: 8542

```
[manalithakar@Manalis-MacBook-Air ~ % ./17q1
```

--- DIGIT OPERATIONS MENU ---

1. Generate Largest Number
2. Generate Smallest Number
3. Swap First and Last Digit
4. Check Palindrome
5. Exit

Enter your choice: 2

Enter a positive integer: 5082

Smallest possible number: 2058

```
[manalithakar@Manalis-MacBook-Air ~ % ./17q1
```

--- DIGIT OPERATIONS MENU ---

1. Generate Largest Number
2. Generate Smallest Number
3. Swap First and Last Digit
4. Check Palindrome
5. Exit

Enter your choice: 3

Enter a positive integer: 7394

Swapped number: 4397

```
[manalithakar@Manalis-MacBook-Air ~ % ./17q1
```

--- DIGIT OPERATIONS MENU ---

1. Generate Largest Number
2. Generate Smallest Number
3. Swap First and Last Digit
4. Check Palindrome
5. Exit

Enter your choice: 4

Enter a positive integer: 12321

12321 is a Palindrome number.

```
manalithakar@Manalis-MacBook-Air ~ % █
```